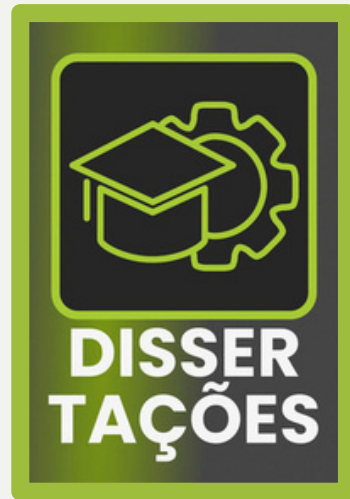




DSD – M4: System component validation (performance, scalability, testing, security)

Projeto em Engenharia
Informática 2025/2026

Presented by
Group 16

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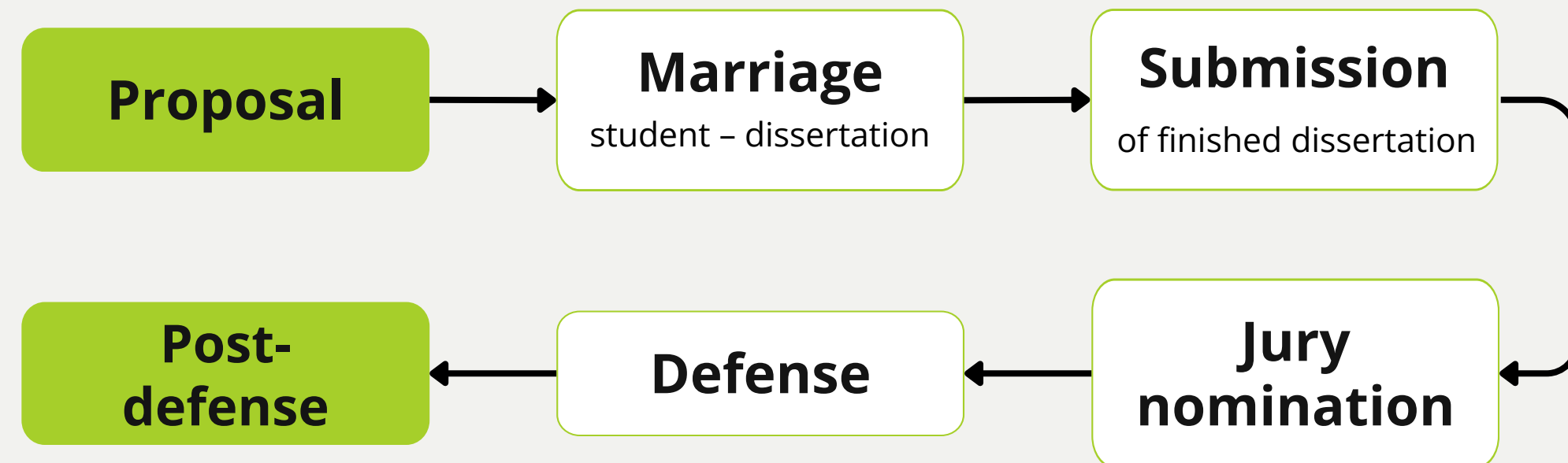
DSD – MS4

Project context / Platform scope

DSD is not only defense management.

It covers the dissertation / project / internship lifecycle.

Current work extends jury, scheduling, status tracking, quotas and post-defense workflows.



Evolution from main to indev

187

commits from main to indev

52

files changed

+5605 / -747

lines changed

~44

merge commits

```
PS DSD M4 evidence > git_diff_shortstat_origin_main_to_indev
52 files changed, 5605 insertions(+), 747 deletions(-)
```

```
PS DSD M4 evidence > git_diff_stat_origin_main_to_indev
.gitignore | 4 +
backend/api/Dockerfile | 7 +-
backend/api/src/endpoints/auth_endpoints.py | 2 +-
backend/api/src/endpoints/diretor_endpoints.py | 106 +-
.../api/src/endpoints/dissertation_endpoints.py | 1014 ++++++
backend/api/src/endpoints/interest_endpoints.py | 10 +
backend/api/src/endpoints/staff_endpoints.py | 57 +-
backend/api/src/endpoints/submission_endpoints.py | 48 +-
backend/api/src/models/defense_date_request.py | 4 +-
backend/api/src/models/dissertation.py | 20 +-
backend/api/src/models/jury.py | 49 +
backend/api/src/models/set_president_request.py | 6 -
backend/api/src/repositories/courses_repo.py | 6 +
backend/api/src/repositories/dissertations_repo.py | 880 ++++++
backend/api/src/repositories/logs_repo.py | 34 +
backend/api/src/startup/api_startup.py | 4 +
backend/api/src/templates/template_ata_defesa.docx | Bin 0 -> 8452 bytes
.../src/templates/template_parecer_individual.docx | Bin 0 -> 8437 bytes
backend/api/src/utils/auth.py | 7 +
backend/api/src/utils/defense_minutes.py | 21 +
backend/api/src/utils/email_notifications.py | 2 +-
backend/api/src/utils/enums.py | 10 +
backend/api/src/utils/individual_opinion.py | 21 +
docker-compose.yml | 10 +-
frontend/Dockerfile | 7 +-
frontend/src/App.jsx | 22 +-
frontend/src/actions/deleteActions.jsx | 8 +-
frontend/src/actions/getActions.jsx | 63 +-
frontend/src/actions/postActions.jsx | 30 +-
frontend/src/actions/putActions.jsx | 19 +-
frontend/src/assets/dsd_bg.png | Bin 1449984 -> 490076 bytes
frontend/src/assets/dsd_bg_big.png | Bin 0 -> 1449984 bytes
frontend/src/components/cards/PresidentCard.jsx | 15 +-
frontend/src/components/layout/Navbar.jsx | 34 +-
frontend/src/components/modals/AliaModal.jsx | 30 +-
.../src/components/modals/ScheduleDefenseModal.jsx | 39 +-
.../components/modals/ScheduleDefenseModalID.jsx | 237 +++++
frontend/src/components/pages/CalendarPage.jsx | 57 +-
.../pages/DiretorDissertationsStatus.jsx | 469 ++++++
.../components/pages/DiretorValidationPage.jsx | 0
.../src/components/pages/DissertationsDetails.jsx | 94 +-
frontend/src/components/pages/GerirPage.jsx | 2 +-
frontend/src/components/pages/JuryQuotasPage.jsx | 353 ++++++
frontend/src/components/pages/LandingPage.jsx | 2 +-
.../src/components/pages/MyPresidenciesPage.jsx | 491 ++++++
.../pages/PresidentWorkspacePlaceholderPage.jsx | 51 +
.../src/components/pages/SubmitFilesPageJury.jsx | 203 ---
frontend/src/components/pages/SubmitJuryPage.jsx | 718 ++++++
.../src/components/pages/SubmitPresidentPage.jsx | 104 --
.../pages/staff/StaffDissertationsStatus.jsx | 530 ++++++
.../components/pages/staff/ValidateDocuments.jsx | 105 +-
.../staff/StaffDefenseScheduleAction.jsx | 347 ++++++
52 files changed, 5605 insertions(+), 747 deletions(-)
```

New work: jury workflow, defense scheduling, quotas and operational status pages.

Post-defense scope: document workflow, minutes/opinions generation, grade registration and presidencies.

Main remains the inherited baseline; indev shows this year's integration work.

Performance validation

Endpoint	Requests	Success	Avg	p50	p95
/dissertation/list	50	50	129.76 ms	122.23 ms	188.93 ms
/staff/dissertations/status	50	50	394.84 ms	379.69 ms	476.83 ms
/dissertation/jury/quotas	50	50	106.74 ms	100.56 ms	149.94 ms
/dissertation/defenses? month=05&year=2026	50	50	20.85 ms	18.12 ms	37.48 ms

```
PS DSD M4 evidence > performance_baseline
Endpoint                                Requests  Success  Avg          p50         p95
/dissertation/list                      50        50       129.76 ms    122.23 ms   188.93 ms
/staff/dissertations/status              50        50       394.84 ms    379.69 ms   476.83 ms
/dissertation/jury/quotas                50        50       106.74 ms    100.56 ms   149.94 ms
/dissertation/defenses?month=05&year=2026 50        50       20.85 ms     18.12 ms    37.48 ms
```

Baseline: 50 sequential requests per endpoint
Metrics: average, p50, p95 and success rate
Local environment, not production hardware

Main finding
The Secretary status endpoint is the heaviest because it builds a richer operational view and crosses dissertation, document and student data.

Load testing with k6

```
PS DSD M4 evidence > k6_load_test_summary
k6 load test summary - DSD M4
Scenario: 5 VUs for 30s
Endpoints: dissertation list, staff status, jury quotas, defenses by month

Overall results
  HTTP requests: 160 (4.98 req/s)
  Iterations: 40 (1.25 iter/s)
  Checks: 160 passed / 0 failed (100.00%)
  HTTP failures: 0.00%

Latency
  avg: 752.10 ms
  median: 357.94 ms
  p90: 1757.45 ms
  p95: 2488.75 ms
  max: 4439.99 ms

Thresholds
  checks rate > 99%: PASSED
  http_req_failed rate < 1%: PASSED
  global p95 < 700 ms: FAILED
```

- Controlled local load test: 5 VUs for 30 seconds.
- Checks validate HTTP success for all critical endpoints.
- Thresholds validate error rate and p95 latency.
- Result: 0% HTTP failures, but global p95 threshold failed under load.

Interpretation

The system stayed available under light/moderate load, while latency exposed bottlenecks for heavier list/status endpoints.

Scalability analysis

FastAPI service can scale vertically, and theoretically horizontally.

MongoDB and MariaDB are central shared dependencies.

Local filesystem for uploaded documents limits multiple API instances.

External dependencies: UA auth, email and signature validation services.

Future mitigation: shared/object storage, indexes and background processing.



PS DSD-M4-evidence > docker_stats_during_benchmark

Command: docker stats --no-stream sampled while running benchmark

Benchmark: py -3.11 scripts/ml_measure_endpoints.py --insecure --requests 100 --warmup 5

Paths: /dissertation/list, /stuff/dissertations/status, /dissertation/jury/quotes

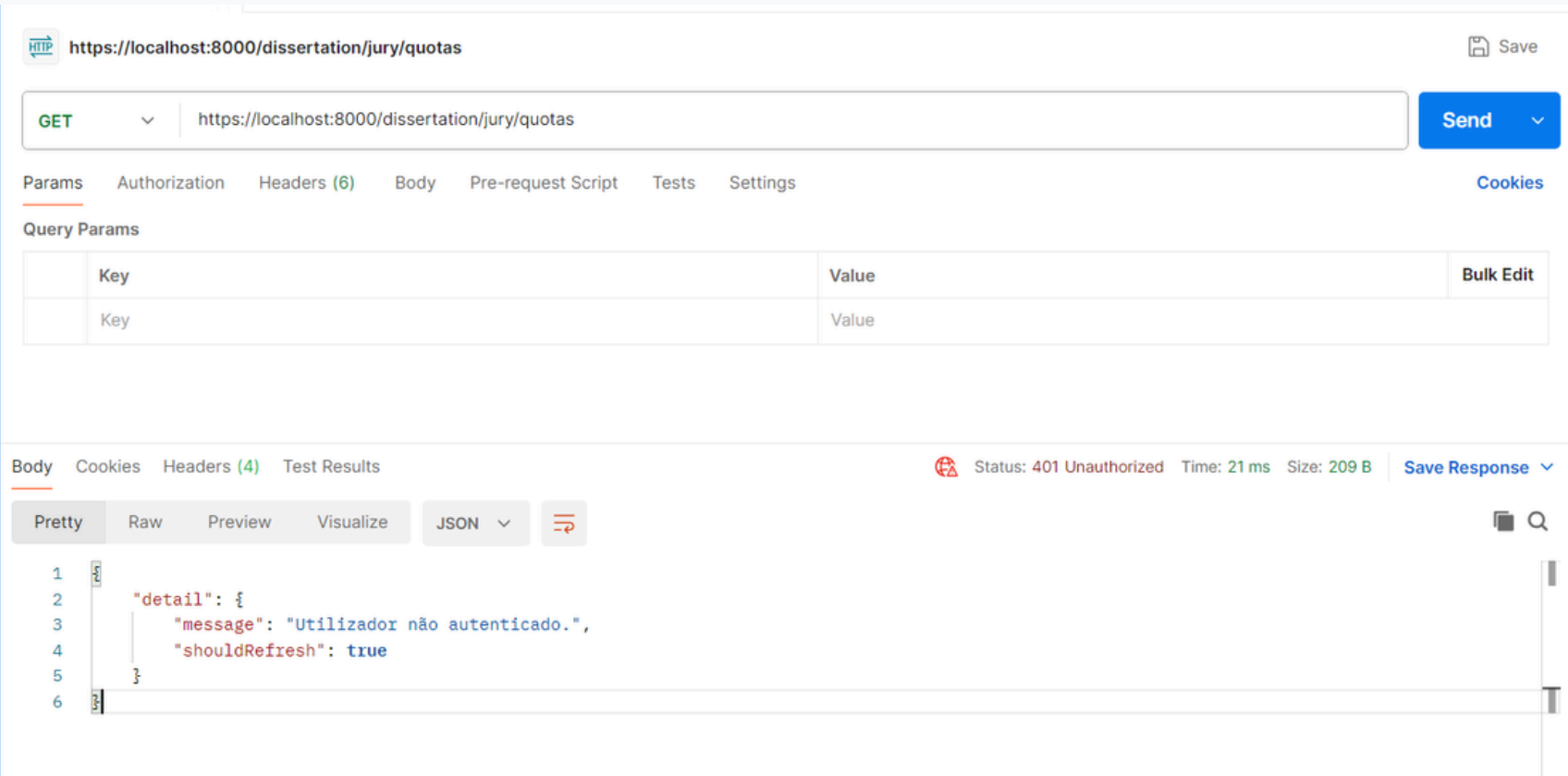
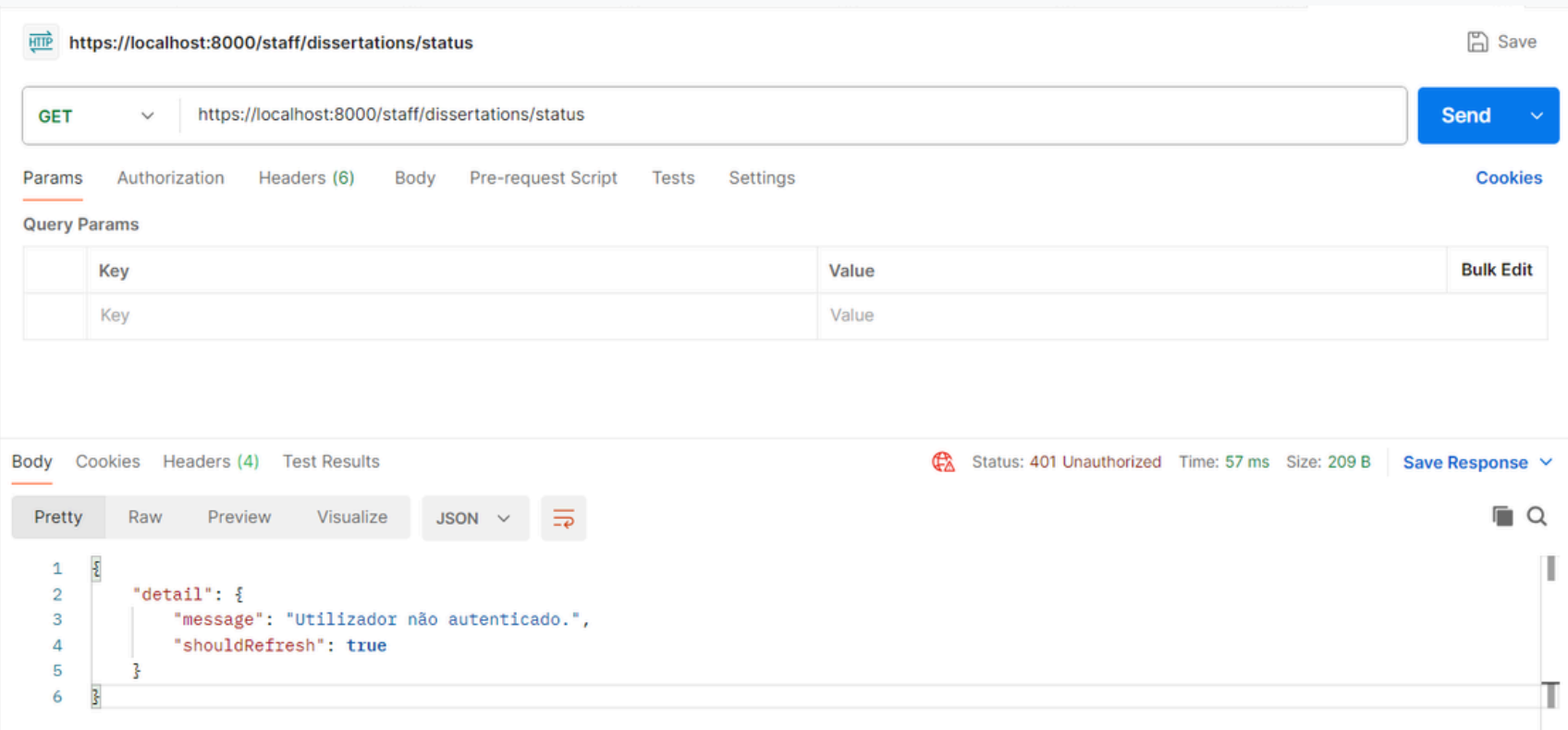
=== sample 1 2026-05-18 01:59:56 ===										
CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS			
2084032dc67	dsd2025-app-1	33.15%	136.8MiB / 7.65GiB	1.75%	6.68MB / 189MB	7.14MB / 0B	23			
4116ae0ccdb	dsd2025-api_instance-1	33.98%	93.84MiB / 7.65GiB	1.20%	137MB / 61.5MB	659KB / 28.7KB	18			
e3dc58b94b5a	dsd2025-mongo_db-1	7.79%	313.1MiB / 7.65GiB	4.00%	12.1MB / 196MB	113MB / 56MB	68			
c72cb33ce5b	dsd2025-mariadb-1	0.01%	122.0MiB / 7.65GiB	1.56%	111KB / 122KB	47.7MB / 36.9KB	11			
=== sample 2 2026-05-18 01:59:59 ===										
CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS			
2084032dc67	dsd2025-app-1	32.48%	136.9MiB / 7.65GiB	1.75%	6.68MB / 189MB	7.14MB / 0B	23			
4116ae0ccdb	dsd2025-api_instance-1	27.16%	93.85MiB / 7.65GiB	1.20%	138MB / 62MB	659KB / 28.7KB	18			
e3dc58b94b5a	dsd2025-mongo_db-1	5.07%	313.4MiB / 7.65GiB	4.00%	12.2MB / 196MB	113MB / 56.1MB	68			
c72cb33ce5b	dsd2025-mariadb-1	0.02%	122.0MiB / 7.65GiB	1.56%	111KB / 122KB	47.7MB / 36.9KB	11			
=== sample 3 2026-05-18 02:00:03 ===										
CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS			
2084032dc67	dsd2025-app-1	32.02%	137.0MiB / 7.65GiB	1.75%	6.68MB / 189MB	7.14MB / 0B	23			
4116ae0ccdb	dsd2025-api_instance-1	31.11%	93.84MiB / 7.65GiB	1.20%	139MB / 62.6MB	659KB / 28.7KB	18			
e3dc58b94b5a	dsd2025-mongo_db-1	6.70%	313.4MiB / 7.65GiB	4.00%	12.3MB / 196MB	113MB / 56.1MB	68			
c72cb33ce5b	dsd2025-mariadb-1	0.01%	122.0MiB / 7.65GiB	1.56%	111KB / 122KB	47.7MB / 36.9KB	11			
=== sample 4 2026-05-18 02:00:07 ===										
CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS			
2084032dc67	dsd2025-app-1	34.48%	136.8MiB / 7.65GiB	1.75%	6.68MB / 189MB	7.14MB / 0B	23			
4116ae0ccdb	dsd2025-api_instance-1	53.38%	94.51MiB / 7.65GiB	1.21%	140MB / 63.3MB	659KB / 28.7KB	18			
e3dc58b94b5a	dsd2025-mongo_db-1	17.97%	314.2MiB / 7.65GiB	4.01%	12.0MB / 201MB	113MB / 56.1MB	68			
c72cb33ce5b	dsd2025-mariadb-1	0.03%	122.0MiB / 7.65GiB	1.56%	111KB / 122KB	47.7MB / 36.9KB	11			
=== sample 5 2026-05-18 02:00:11 ===										
CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS			
2084032dc67	dsd2025-app-1	31.80%	137.2MiB / 7.65GiB	1.75%	6.68MB / 189MB	7.14MB / 0B	23			
4116ae0ccdb	dsd2025-api_instance-1	72.81%	94.39MiB / 7.65GiB	1.20%	156MB / 69.6MB	659KB / 28.7KB	18			
e3dc58b94b5a	dsd2025-mongo_db-1	20.42%	314.2MiB / 7.65GiB	4.01%	13MB / 210MB	113MB / 56.1MB	68			
c72cb33ce5b	dsd2025-mariadb-1	0.01%	122.0MiB / 7.65GiB	1.56%	111KB / 122KB	47.7MB / 36.9KB	11			
=== sample 6 2026-05-18 02:00:14 ===										
CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS			
2084032dc67	dsd2025-app-1	34.34%	136.1MiB / 7.65GiB	1.74%	6.68MB / 189MB	7.14MB / 0B	23			
4116ae0ccdb	dsd2025-api_instance-1	72.96%	94.16MiB / 7.65GiB	1.20%	164MB / 73.8MB	659KB / 28.7KB	18			
e3dc58b94b5a	dsd2025-mongo_db-1	17.91%	314.2MiB / 7.65GiB	4.01%	13.9MB / 225MB	113MB / 56.1MB	68			
c72cb33ce5b	dsd2025-mariadb-1	0.01%	122.0MiB / 7.65GiB	1.56%	111KB / 122KB	47.7MB / 36.9KB	11			
=== sample 7 2026-05-18 02:00:18 ===										
CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS			
2084032dc67	dsd2025-app-1	30.20%	136.9MiB / 7.65GiB	1.75%	6.68MB / 189MB	7.14MB / 0B	23			
4116ae0ccdb	dsd2025-api_instance-1	72.41%	93.89MiB / 7.65GiB	1.20%	170MB / 70MB	659KB / 28.7KB	18			
e3dc58b94b5a	dsd2025-mongo_db-1	17.11%	314.2MiB / 7.65GiB	4.01%	14.5MB / 238MB	113MB / 56.1MB	68			
c72cb33ce5b	dsd2025-mariadb-1	0.01%	122.0MiB / 7.65GiB	1.56%	111KB / 122KB	47.7MB / 36.9KB	11			
=== sample 8 2026-05-18 02:00:22 ===										
CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS			
2084032dc67	dsd2025-app-1	27.82%	136.9MiB / 7.65GiB	1.75%	6.68MB / 189MB	7.14MB / 0B	23			
4116ae0ccdb	dsd2025-api_instance-1	47.69%	94.15MiB / 7.65GiB	1.20%	188MB / 83.3MB	659KB / 28.7KB	18			
e3dc58b94b5a	dsd2025-mongo_db-1	14.04%	314.2MiB / 7.65GiB	4.01%	15MB / 247MB	113MB / 56.2MB	68			
c72cb33ce5b	dsd2025-mariadb-1	0.01%	122.0MiB / 7.65GiB	1.56%	111KB / 122KB	47.7MB / 36.9KB	11			
=== sample 9 2026-05-18 02:00:26 ===										
CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS			
2084032dc67	dsd2025-app-1	33.81%	136.5MiB / 7.65GiB	1.74%	6.68MB / 189MB	7.14MB / 0B	23			
4116ae0ccdb	dsd2025-api_instance-1	68.33%	94.15MiB / 7.65GiB	1.20%	201MB / 80MB	659KB / 28.7KB	18			
e3dc58b94b5a	dsd2025-mongo_db-1	18.06%	314.2MiB / 7.65GiB	4.01%	15.6MB / 258MB	113MB / 56.2MB	68			
c72cb33ce5b	dsd2025-mariadb-1	0.07%	122.5MiB / 7.65GiB	1.56%	111KB / 122KB	47.7MB / 36.9KB	9			
=== sample 10 2026-05-18 02:00:29 ===										
CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS			
2084032dc67	dsd2025-app-1	32.23%	137MiB / 7.65GiB	1.75%	6.68MB / 189MB	7.14MB / 0B	23			
4116ae0ccdb	dsd2025-api_instance-1	73.58%	93.91MiB / 7.65GiB	1.20%	209MB / 92.3MB	659KB / 28.7KB	18			
e3dc58b94b5a	dsd2025-mongo_db-1	19.98%	314.2MiB / 7.65GiB	4.01%	16.2MB / 269MB	113MB / 56.2MB	68			
c72cb33ce5b	dsd2025-mariadb-1	0.01%	122.5MiB / 7.65GiB	1.56%	111KB / 122KB	47.7MB / 36.9KB	9			
=== sample 11 2026-05-18 02:00:33 ===										
CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS			

Security validation

- Protected routes require authentication.
- Role-based access appears across student, teacher, staff, admin and Diretor-* flows.
- Invalid or missing access returns controlled 401/403 responses.
- Documents and uploads remain the most sensitive access-control surface.

Risk focus

RBAC is effective in key endpoints, but rules are dispersed and should be consolidated/tested.



Business invariants validation

Invariant	Test	Expected	Status
Grade must be 0-20	grade=21	422	Passed
Defense requires valid date	past date	400	Passed
Override requires justification	override without reason	400	Passed
Minutes require defense data	missing date/room/jury	400	Passed
Workflow cannot skip invalid states	repo transition checks	reject invalid transition	Designed

```
PS DSD M4 evidence > grade_out_of_range_422
HTTP/1.1 422 Unprocessable Entity
date: Mon, 18 May 2026 00:52:07 GMT
server: uvicorn
content-length: 189
content-type: application/json

{"detail":[{"type":"less_than_equal","loc":["body"],"msg":"Input should be less than or equal to 20","input":21,"ctx":{"le":20,"url":"https://errors.pydantic.dev/2.12/v/less_than_equal"}]}
```

```
PS DSD M4 evidence > override_without_reason_400
HTTP/1.1 400 Bad Request
date: Mon, 18 May 2026 00:52:08 GMT
server: uvicorn
content-length: 82
content-type: application/json

{"detail":"Ao fazer override de um conflito, é obrigatória uma justificação."}
```

These tests exercise backend validation paths and expected error responses.

Concurrency and consistency risks

No transactions:

- Two staff members can try to schedule the same room/time.
- Two users can upload/update the same signed document.
- A file may be saved while Mongo metadata update fails.

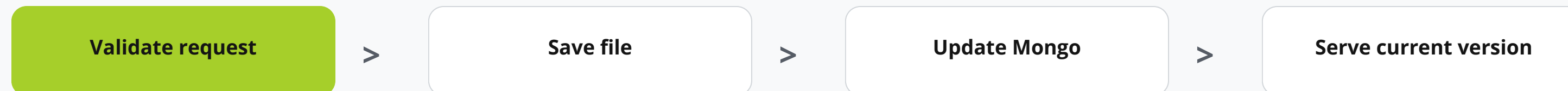
Current code relies mostly on simple update_one operations, not transactions.
Mitigations: conditional updates, versioned paths, cleanup jobs and targeted tests.

Consistency gap

The critical boundary is between filesystem writes and Mongo updates. If only one side succeeds, the visible document state may become inconsistent.

Validation target

Concurrent scheduling and uploads should be tested with repeated parallel requests.



Testing status and gaps

- Automated test coverage is still limited in the inherited codebase.
- M4 evidence combines build/lint/compile checks with manual endpoint validation.
- Missing automated tests: RBAC, workflows, uploads, quotas and defense conflicts.
- Next step: automate critical backend workflow and permission tests.

Git diff check: PASS

✖428 problems (398 errors, 30 warnings)
6 errors and 0 warnings potentially fixable with the `--fix` option.

```
PS C:\Users\rafae\OneDrive\Ambiente de Trabalho\rafa\2º Semestre\PEI\dsd2025> py -3.11 -m compileall -q backend/api/src
>> if ($LASTEXITCODE -eq 0) { "Backend py_compile: PASS" } else { "Backend py_compile: FAIL" }
Backend py_compile: PASS
```

```
PS C:\Users\rafae\OneDrive\Ambiente de Trabalho\rafa\2º Semestre\PEI\dsd2025\frontend> npm.cmd run build

> frontend@0.0.0 build
> vite build

vite v5.4.18 building for production...
transforming (4) node_modules\@tanstack\react-query\build\modern\index.jsBrowserslist: browsers data (caniuse-lite) is 13 months old. Please run:
  npx update-browserslist-db@latest
  Why you should do it regularly: https://github.com/browserslist/update-db#readme

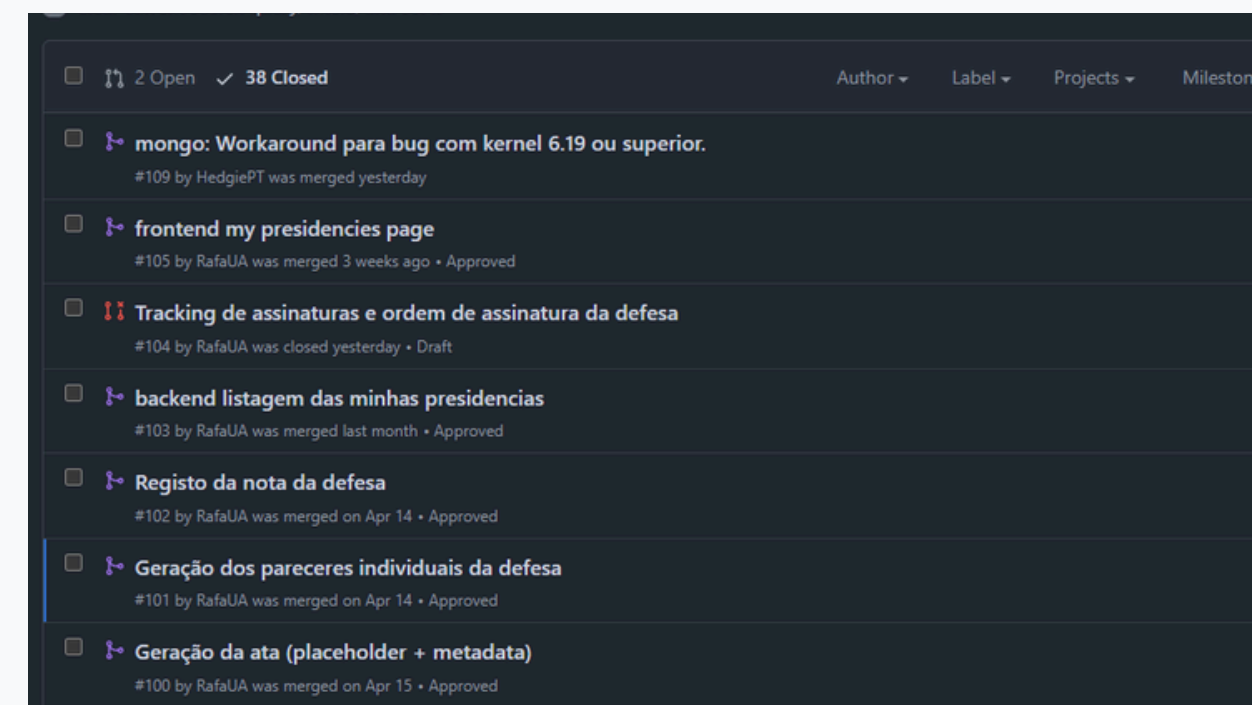
📦 daisyUI 4.12.24
  ✓ 2 themes added      https://daisyui.com/docs/themes
  ★ Star daisyUI on GitHub https://github.com/saadeghi/daisyui

transforming (5) node_modules\react-cookie\esm\index.mjs
/fonts/roboto-v27-latin-regular.eot referenced in /fonts/roboto-v27-latin-regular.eot didn't resolve at build time, it will remain unchanged to be resolved at runtime
✓ 816 modules transformed.
dist/index.html                                0.56 kB | gzip: 0.35 kB
dist/assets/icon_text_v2-DotGj2yi.svg          8.10 kB | gzip: 3.25 kB
dist/assets/icon_text-LfFsw6E9.svg             8.27 kB | gzip: 3.24 kB
dist/assets/icon-gv5DEIvg.svg                 14.71 kB | gzip: 5.92 kB
dist/assets/deti-DoEttNBB.png                  16.01 kB
dist/assets/icon_xl_text_vertical-Cp47giFS.svg 22.59 kB | gzip: 9.13 kB
dist/assets/icon_xl_text_vertical-Cp47giFS.svg 23.46 kB | gzip: 9.53 kB
dist/assets/icon_md_text-Q45_kkXx.svg          27.78 kB | gzip: 10.86 kB
dist/assets/icon_xs_text-BqdItLNB.svg          27.99 kB | gzip: 11.00 kB
dist/assets/deti_logo_old-Rowv-sLW.svg         29.36 kB | gzip: 10.47 kB
dist/assets/missing-dYiNCM9d.svg              32.16 kB | gzip: 8.74 kB
dist/assets/user-DZ-nKL20.png                  40.37 kB
dist/assets/dsd_bg-DgJ0IRss.png               490.08 kB
dist/assets/index-D7HuTLvi.css                149.35 kB | gzip: 20.97 kB
dist/assets/index-nD1DnonJ.js                 1,126.31 kB | gzip: 334.71 kB

(!) Some chunks are larger than 500 kB after minification. Consider:
- Using dynamic import() to code-split the application
- Use build.rollupOptions.output.manualChunks to improve chunking: https://rollupjs.org/configuration-options/#output-manualchunks
- Adjust chunk size limit for this warning via build.chunkSizeWarningLimit.
✓ built in 10.88s
```

Project management

- Gestão com GitHub: issues, milestones, labels e PRs
- Tarefas divididas por frontend, backend, validação, segurança e fluxos administrativos
- 23 issues abertas e 40 PRs no total
- Algumas tarefas transitaram por maior complexidade técnica
- Docker/scripts ajudaram, mas faltam testes automáticos e CI/CD mais robusto



Next steps

Stress tests

Run larger tests with seeded datasets and realistic concurrency.

RBAC tests

Automate student/teacher/staff/admin negative access cases.

Permissions

Consolidate permission checks around sensitive document endpoints.

Storage

Move uploaded documents toward shared/object storage for scaling.

CI quality gates

Add tests, lint, encoding checks and dependency audit to CI.

Priority: turn the M4 evidence into repeatable checks, then extend coverage around high-risk workflows.

Thank you!



 <https://pei-dsd-2025.github.io/microsite/> 